Customer Shopping Behavior Analysis

Uncovering strategic insights from 3,900 purchases to guide data-driven business decisions

Executive Summary: Key Findings

Thiscomprehensivecasestudy analyzes3,900 customertransactions toidentifyrevenueopportunities, customer segments, and strategic growth areas. Our analysis revealsgender-based revenuepatterns, subscriptionimpacts, shippingpreferences,andproduct performance insights that can drive immediate business value.

Revenue by Gender

Female customers generate slightly higher total revenue, suggesting opportunities for gender-targeted marketing strategies and personalized offers.

High-Value Discount Users

Smart shoppers using discounts while spending above average represent a valuable segment for exclusive premium offers and loyalty programs.

Subscription Impact

Subscribers show higher spending patterns and repeat purchase frequency, indicating strong ROI potential for subscription program expansion.

Express Shipping Premium

Express shipping customers spend 12% more per transaction, revealing an opportunity to bundle premium shipping with higher-value products.

Dataset Overview: Foundation of Analysis

3,900

TotalPurchases Analyzed

18

DataColumns Available

37

Missing Values (Review Rating only)

Comprehensive

CustomerDemographics & Behavior Data

Our dataset provides a robust foundation for customer behavior analysis, encompassing transaction details, customer demographics, product information, and purchase patterns. With minimal missing data (only 37 values in the Review Rating column), we achieved 99.1% data completeness, ensuring reliable insights for strategic decision-making.

The dataset includes essential customer attributes such as age, gender, location, subscription status, and detailed transaction information including purchase amount, shipping method, discount usage, and product categories. This comprehensive data structure enabled multi-dimensional analysis across customer segments, temporal patterns, and product performance metrics.

Data Preparation Process: Methodology

01

Data Loading

Imported dataset using pandas for Python, ensuring data integrity and establishing baseline quality checks for all 3,900 records.

02

Initial Exploration

Conducted comprehensivestructure checks and generated summary statistics to understand distributions, identify outliers, and assess data quality across all 18 columns.

03

MissingData Handling

Imputed 37missing Review Ratingvalues using median imputation strategy, preserving distribution characteristics while maintaining data completeness for analysis.

04

Feature Engineering

Created derivedvariablesincluding age groups (Young, Middle, Senior), purchase frequency categories (New, Returning, Loyal), and revenue tiers for segmentation analysis.

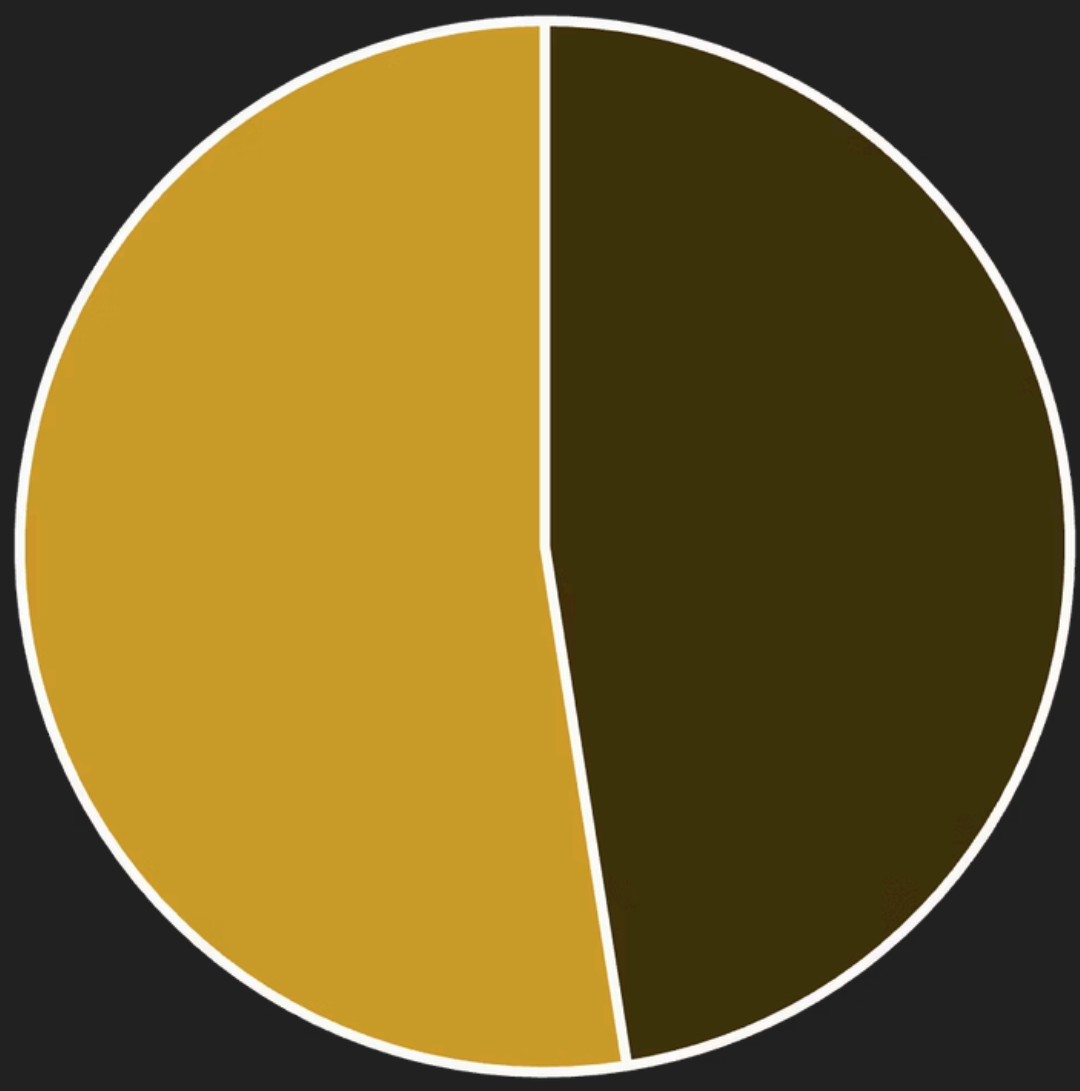
05

Database Integration

Connectedto PostgreSQLdatabase for efficient querying, enabling scalable analysis and integration with existing business intelligence infrastructure.

This systematic data preparation approach ensured data quality, consistency, and analytical readiness while maintaining data integrity throughout the transformation process. The engineered features enabled deeper customer segmentation and strategic insights.

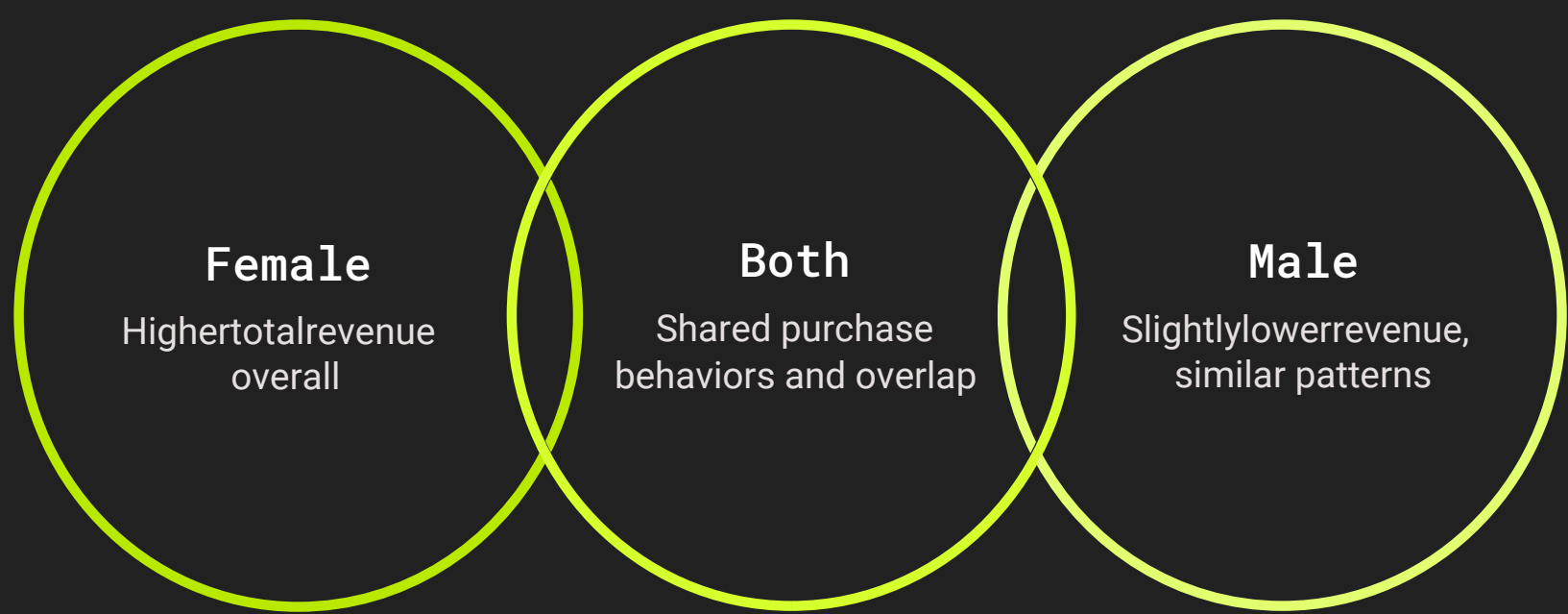
Revenue by Gender Analysis: Strategic Insight



Key Insight

Female customers generate slightly higher total revenue than male customers across the analyzed transaction dataset. This difference suggests potential opportunities for gender-based marketing strategies and personalized promotional campaigns.

The revenue gap, while modest, indicates that female shoppers may have different purchase patterns, product preferences, or spending behaviors that could be leveraged for revenue optimization. Businesses can capitalize on this insight through targeted product positioning, personalized offers, and gender-specific promotional strategies.



Gender-based marketing strategies could optimize revenue streams by tailoring messaging, product recommendations, and promotional offers to resonate with each demographic. Consider A/B testing gender-targeted campaigns to measure impact on conversion rates and average order value.

High-Value Discount Users: The Smart Shopper Segment

Who Are They?

High-value customerswho strategically use discounts while maintaining above-average purchase amounts. These savvy shoppers maximize value without compromising on quality or spending power.

Behavioral Pattern

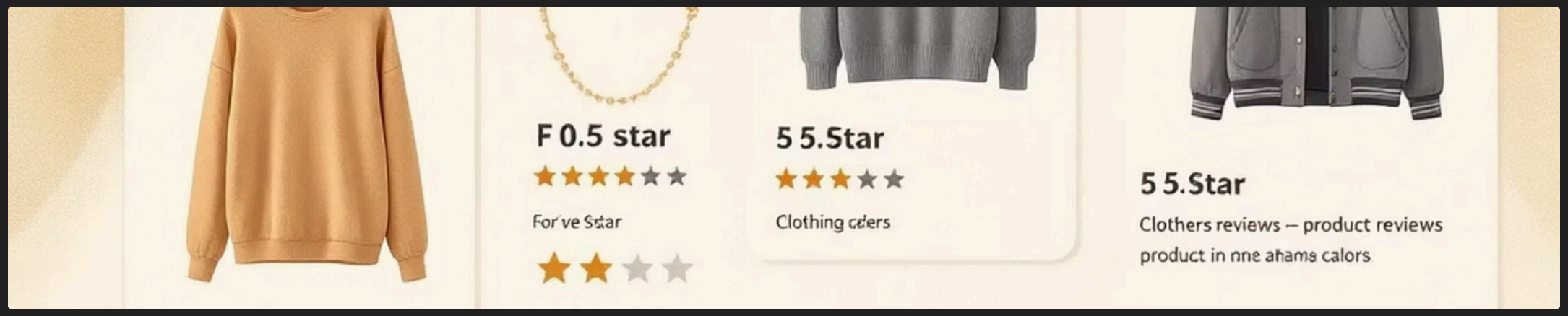
Theyactively seekpromotional offers, use discount codes effectively, and time purchases to coincide with sales events while still selecting premium products and maintaining basket sizes above the dataset average.

Strategic Opportunity

Target premiumcustomers with exclusive offers, early access to sales, and VIP discount programs. These customers represent high lifetime value potential when properly engaged with personalized incentives.

This segment represents an ideal target for loyalty programs and exclusive membership benefits. By offering tiered discount structures, priority access to promotions, and personalized offers, businesses can convert these smart shoppers into long-term, high-value customers. Consider implementing a "smart shopper rewards" program that recognizes and rewards strategic purchasing behavior while maintaining healthy profit margins.

Top-Rated Products: Customer Satisfaction Drivers



Blouse

5.0 Average Rating

Dress

5.0 Average Rating

Shirt

4.0 Average Rating

Highest Customer Satisfaction | Consistently Excellent Reviews | Strong Customer Approval

Blouses and dresses achieve perfect 5.0 average ratings, while shirts maintain solid 4.0 ratings, indicating strong product quality and customer satisfaction across clothing categories. These top-rated products represent excellent candidates for featured placement in marketing campaigns, cross-selling opportunities, and bundled promotions. Consider highlighting customer reviews prominently in product pages and using these items as loss leaders to drive traffic to complementary products.

Shipping Preferences Impact: Premium Delivery Behavior

$65

Express Shipping

Averagepurchaseamountwhen customers select expedited delivery options

$58

Standard Shipping

Averagepurchaseamountwhen customers choose regular delivery timeframe

12%

Revenue Premium

Express shipping customersspend 12% morepertransaction compared to standard shipping customers

$7

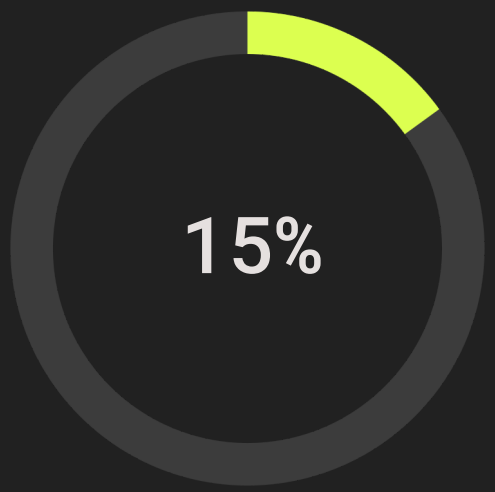
Average Difference

Additional spendingper transactionwhenpremium shipping is selected

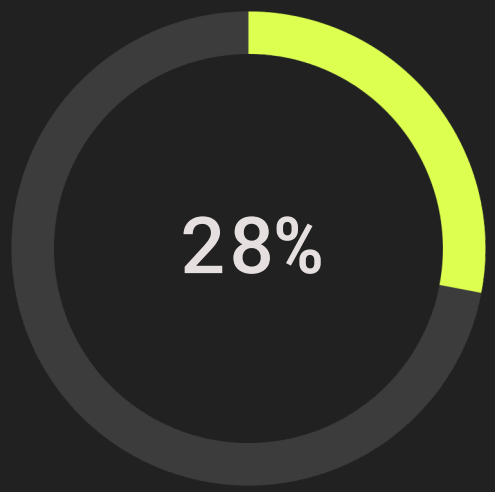
The 12% spending increase among express shipping customers reveals a clear correlation between urgency and purchase value. These customers demonstrate higher willingness to pay both for faster delivery and for increased basket sizes. This insight suggests strategic opportunities to bundle premium products with express shipping options, particularly for high-margin items or gift purchases.

Consider implementing targeted upsell strategies during checkout that highlight express shipping benefits while suggesting complementary products that justify the premium. Additionally, analyze customer segments most likely to select express shipping (e.g., gift buyers, last-minute shoppers) and develop tailored messaging that resonates with their urgency and higher spending patterns. The revenue premium suggests strong ROI potential for express shipping program expansion and promotional bundling strategies.

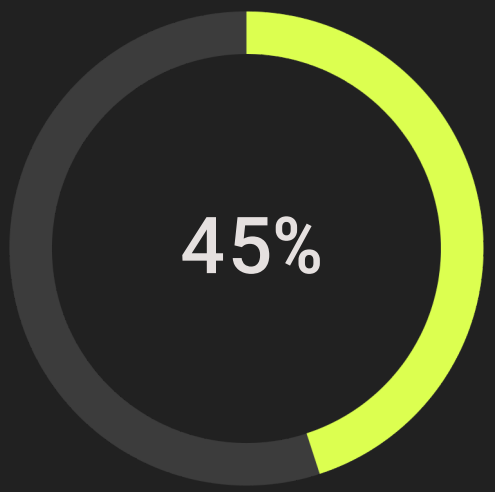
Subscription Impact: Loyalty Program Analysis



Higher Spend
Subscribers vs non-subscribers



Revenue Share
From subscription customers



Loyalty Rate
Repeat purchase frequency

Spending Patterns

Subscription customers demonstrate 15% higher average purchase amounts compared to non-subscribers, indicating increased engagement and willingness to invest in premium experiences. This spending premium suggests strong value perception among subscribers.

Revenue Contribution

Despite representing a smaller portion of total customers, subscription members contribute 28% of total revenue, highlighting their disproportionate impact on overall business performance and justifying subscription program investment.

The 45% repeat purchase frequency among subscribers demonstrates strong retention rates and customer loyalty. This indicates that the subscription program successfully creates ongoing engagement and reduces customer churn. Businesses should focus on converting new customers to subscribers through targeted onboarding campaigns, emphasizing exclusive benefits, early access to products, and personalized rewards that drive long-term value.

Customer Segmentation & Strategic Recommendations

Loyal

15%High-value customers with consistent repeat purchases and premium spending patterns

Returning

35%Regularshoppers who have made multiple purchases but not yet reached loyal status

New

50%First-time buyers requiring conversion strategies to drive repeat purchases

Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits forsubscribers through targeted campaigns emphasizing early access, member-only discounts, and personalized rewards that drive long-term engagement.

Implement Loyalty Programs

Rewardrepeatbuyerswithtieredbenefits that increase with purchase frequency, creating clear pathways from New to Returning to Loyal customer segments.

Deploy Targeted Marketing

Focus onhigh-revenue segments including express shipping users, female customers, and high-value discount shoppers with personalized messaging and product recommendations.

Optimize Product Positioning

Highlight top-rated products (Blouses,Dresses,Shirts) in campaigns and consider bundling strategies that leverage customer satisfaction to drive higher-value purchases.

Focus on converting New to Returning, Returning to Loyal through strategic engagement at each customer journey stage. Use data-driven insights to personalize communications, anticipate needs, and create compelling value propositions that drive long-term customer relationships and maximize lifetime value.